

## **Problem Statement: AI-Driven Cost Estimation for Mechanical Part Manufacturing**

In the mechanical manufacturing industry, accurate and timely cost estimation for part production is a critical yet complex task. Traditionally, this process relies heavily on manual interpretation of engineering drawings, domain expertise, and fragmented costing methods. This leads to inconsistencies, time delays, dependency on skilled estimators, and potential inaccuracies in quotation and pricing.

The objective of this project is to develop an intelligent, AI-ML powered software system that automates the cost estimation process using engineering drawing PDFs as the primary input. The system will be capable of extracting critical design and manufacturing parameters directly from the drawings, including dimensions, tolerances, material specifications, and geometric features.

Based on the extracted data, the software will systematically compute:

- **Raw Material Cost** by determining optimal material size, type, and utilization
- **Machining Cost** by identifying required manufacturing operations (turning, milling, drilling, etc.) and estimating cycle times
- **Manpower Cost** based on labour requirements and process complexity
- **Overheads and Miscellaneous Costs** including tooling, energy consumption, and indirect expenses
- **Logistics and Transportation Cost** based on delivery parameters
- **Profit Margins** as per configurable industry standards

The system will be customizable to specific industry requirements, enabling users to define cost parameters, machining rates, and business rules. Upon uploading a machining drawing PDF, the software will automatically generate a comprehensive and accurate cost breakdown, reducing human intervention and improving efficiency.

This solution aims to enhance decision-making, standardize costing practices, reduce estimation time, and improve competitiveness in the mechanical manufacturing sector through the integration of Artificial Intelligence and Machine Learning.